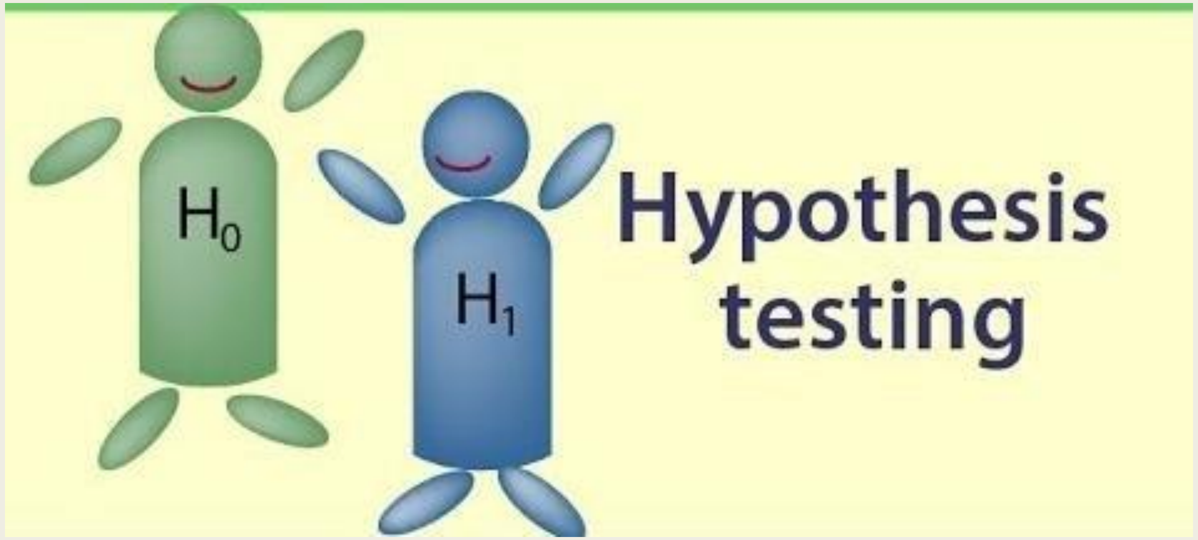


HYPOTHESIS TESTING AND T-TESTS

With Applications in R and SAS

Malith Premarathna

Who we are ?

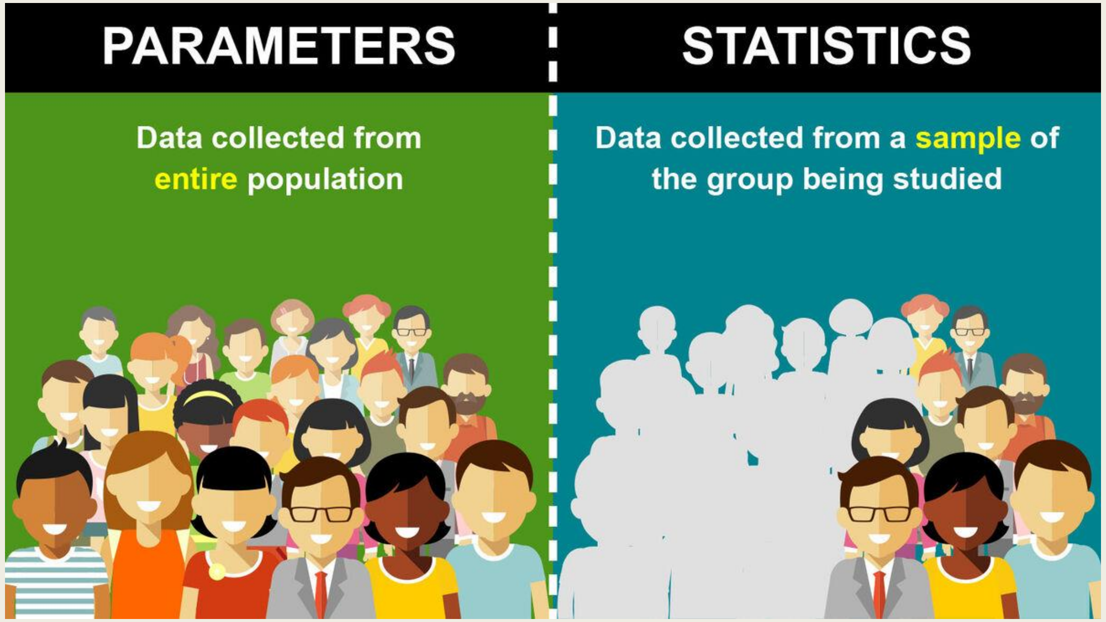
The SC3L is a free service available to students, faculty, and staff at the University of Nebraska–Lincoln who need assistance with a Master's thesis, a PhD dissertation, or faculty research.

You will see today.....

- What is a Hypothesis?
- What is a P-Value?
- Hypothesis Testing Steps
  - *5 main Steps*
- Examples using R
- Examples using SAS

What is Hypothesis?

A **hypothesis** is a statement about a **population parameter** that we **test using sample data**.

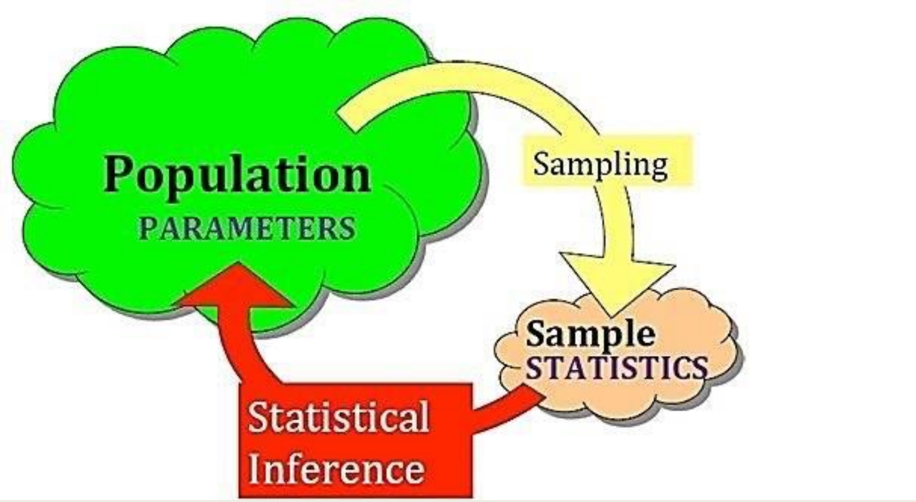


What is Hypothesis?

A **hypothesis** is a statement about a population parameter that we **test using sample data**.

Examples of Parameters and Statistics

Parameter	Statistic
The mean monthly income of your target market	The mean monthly income of 60 of your regular customers
The average age of your target market	The average age of your 100 regular customers



What is Hypothesis?

A **hypothesis** is a statement about a population parameter that we **test using sample data**.

Hypothesis	Meaning
Null Hypothesis ($\boldsymbol{H_0}$)	No effect, No Difference (Default Assumption)
Alternative Hypothesis ($\boldsymbol{H_A}$)	There is an Effect, Difference.

What is a P-Value?

A **p-value** is the probability of obtaining a test statistic **as extreme or more extreme** than the observed one, **assuming that the null hypothesis is true**.

The **p-value** tells us how likely we are to observe our sample results **if H_0 were true**.

Simple Interpretation:

- **Small p-value ($p < 0.05$)** → Unlikely that H_0 is true → **Reject H_0**
- **Large p-value ($p > 0.05$)** → Data is consistent with H_0 → **Fail to reject H_0**

The 5% could be changed to another value.

- called: **Level of Significance**
- Symbol: $\boldsymbol{\alpha}$
- Represents: **% of possibilities we call unusual**

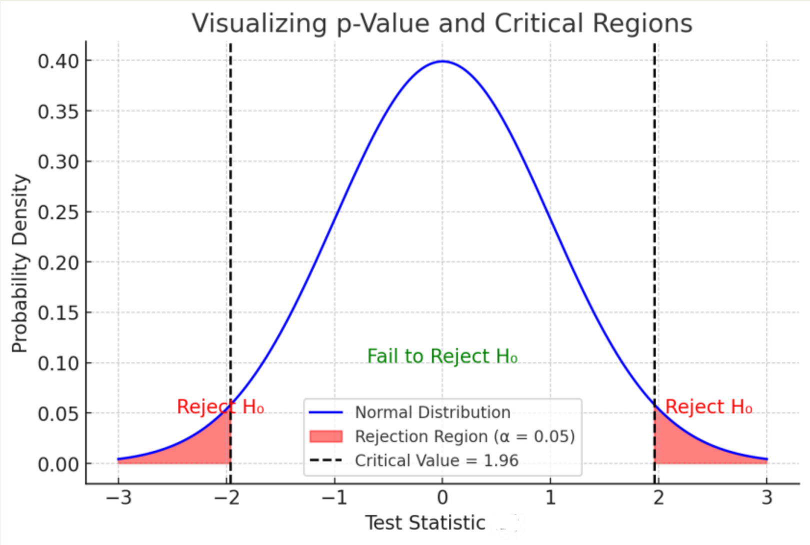
The smaller the p-value, the stronger the evidence against the null hypothesis.

What is a Test Statistic?

A **test statistic** is a numerical value that measures how far **sample data** deviates from a **null hypothesis**

How to Interpret This?

- If the test statistic **falls in the red rejection region**, we **reject H_0**
- If the test statistic **falls in the middle region**, we **fail to reject H_0**



1️⃣ General Formula for a Test Statistic

$$\text{Test Statistic} = \frac{\text{Observed Value} - \text{Expected Value}}{\text{Standard Error}}$$

Where:

- **Observed Value** = Sample statistic (e.g., sample mean, sample proportion).
- **Expected Value** = Hypothesized population parameter under H_0.
- **Standard Error (SE)** = Measures variability in the sample.

Hypotheses testing Steps

- Step 1 : State the Hypotheses $(H_0 \ and \ H_A)$
- Step 2 : Set the Significance Level (α)
- Step 3 : Choose the correct test (t-test)
- Step 4 : Compute the Test Statistics & P Value
- Step 5 : Make a Conclusion (Reject or Fail to Reject H_0)

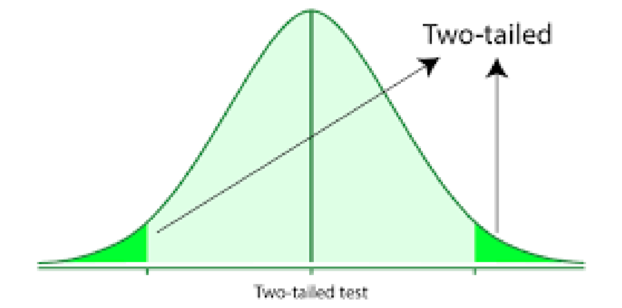
Hypotheses testing Steps

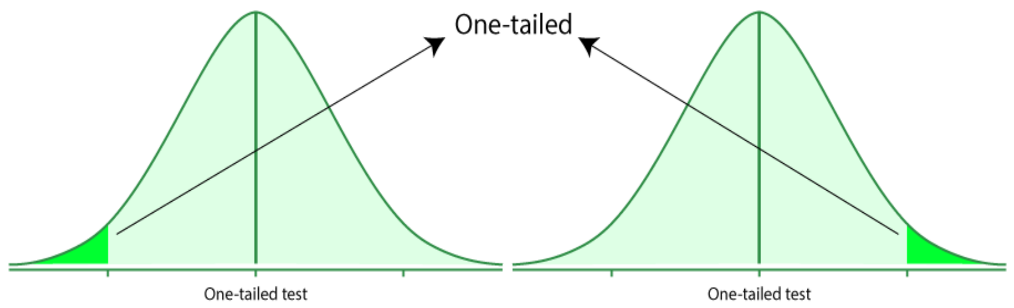
Step 1 : State the Hypotheses (H_0 and H_A)

In general, take the form:

Two-tailed:	Right-tailed:	Left-tailed:
$H_0: \mu = c$	$H_0: \mu \leq c$	$H_0: \mu \geq c$
$H_a: \mu \neq c$	$H_a: \mu > c$	$H_a: \mu < c$

Which H_a depends on research question.





Hypotheses testing Steps

Step 2 : Set the Significance Level (α)

The 5% could be changed to another value.

- called: **Level of Significance**
- Symbol: $\boldsymbol{\alpha}$
- Represents: **% of possibilities we call unusual**

Hypotheses testing Steps

Step 3 : Choose the correct test (t-test)

Situation 1: One Sample t-Test

When to Use?

✅ You are comparing **one sample mean** to a known **population mean**.

🔷 Example 1: Testing Average Weight of Apples

> A farmer claims that his apples weigh **on average 200g**. You collect a sample of **30 apples** and test if their mean weight is significantly different from 200g.

🔷 Example 2: Checking Machine Accuracy

> A soda-filling machine is **supposed** to fill bottles with **500ml** of soda. You take a **random sample** of 50 bottles and test if the mean amount of soda differs from **500ml**.

🔷 Example 3: Exam Scores Compared to a Standard

> The national average **SAT score** is **1050**. A school wants to check if their students' average SAT score **is different** from this national average.

Hypotheses testing Steps

Step 3 : Choose the correct test (t-test)

Situation 2: Two Sample t-Test (Independent Samples)

When to Use?

✅ You are comparing **two different, independent groups.**

🔷 **Example 1: Testing Effect of a New Drug**

> A **new drug** is tested for reducing blood pressure. You randomly assign **50 people** to take the drug and **50 people** to take a placebo. You compare the **average blood pressure** between the two groups.

🔷 **Example 2: Comparing Male vs. Female Running Speeds**

> A researcher wants to compare the **average sprint speed** of **male and female athletes** to see if there is a significant difference.

🔷 **Example 3: A/B Testing for Website Design**

> A company tests **two different website designs** (A & B) and wants to compare the **average time spent** on the site by visitors.

Hypotheses testing Steps

Step 3 : Choose the correct test (t-test)

Situation 3: **Paired t-Test (Dependent Samples)**

When to Use?

✅ You are comparing the **same subjects** before and after treatment.

🔷 **Example 1: Testing a Weight Loss Program**

> A group of **20 people** follows a **6-week diet**. Their weight is recorded **before and after the program**, and we test if there is a **significant weight change**.

🔷 **Example 2: Students' Test Performance Before and After Studying**

> Students take a **math test before and after** using a new learning app. We check if their **scores improved significantly**.

🔷 **Example 3: Effect of Sleep on Reaction Time**

> A researcher measures **reaction time** in **10 participants before and after** getting **8 hours of sleep**.

Hypotheses testing Steps

Step 3 : Choose the correct test (t-test)

Summary of Selecting the Right Test

📌 Summary of Step 3 (Choosing the Right Test)

Type of Data	Test to Use	Example
1 sample vs. population mean	One-Sample t-Test	Test if a machine fills soda cans with 500ml on average
2 independent groups	Two-Sample t-Test	Test if men and women have different average heights
Same group before & after treatment	Paired t-Test ✅	Test if coffee reduces reaction time in the same individuals

Hypotheses testing Steps

Step 3 : Choose the correct test (t-test)

Assumptions of T-Test

Assumption	Explanation
Normality	Data should be normally distributed (for small samples).
Independence	Observations must be independent (no repeated measures in independent t-test).
Equal Variances (Homoscedasticity)	Variance should be equal in both groups (for two-sample t-tests).
Continuous Data	The dependent variable must be continuous (interval/ratio scale).

Hypotheses testing Example 1

Situation 1: One Sample t-Test

🔷 Scenario: Apple Weights vs. 200g

A farmer claims that his apples weigh **200g** on average. We collect a **random sample of 6 apples** and get the following weights (in grams):

$$X = \{198, 202, 205, 197, 200, 204\}$$

We want to test whether the mean weight **differs significantly** from **200g** at **α = 0.05**.

Hypotheses testing Example 1

Situation 1: One Sample t-Test

Step 1: Define Hypotheses

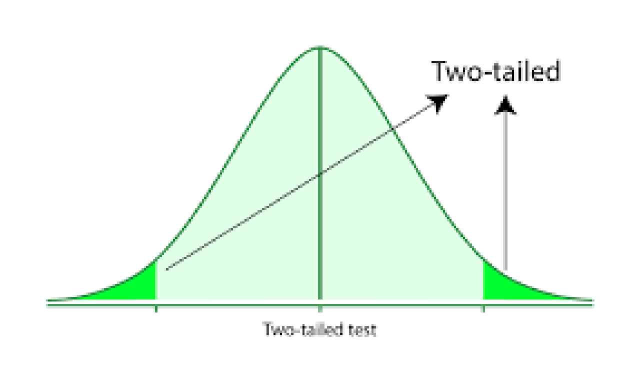
- **Null Hypothesis (H_0)**: The mean weight is **200g**.

$$H_0 : \mu = 200$$

- **Alternative Hypothesis (H_1)**: The mean weight **is not** 200g.

$$H_1 : \mu \neq 200$$

- **Significance level (α)**: **0.05** (two-tailed test).



Hypotheses testing Example 1

Situation 1: One Sample t-Test

Step 2: Calculate Sample Statistics

1 **Sample Mean ($\bar{X}$):**

$$\bar{X} = \frac{198 + 202 + 205 + 197 + 200 + 204}{6} = \frac{1206}{6} = 201$$

2 **Sample Standard Deviation (s):**

The formula for standard deviation:

$$s = \sqrt{\frac{\sum (X_i - \bar{X})^2}{n - 1}}$$

Let's compute $(X_i - \bar{X})^2$:

X_i	$X_i - \bar{X}$	$(X_i - \bar{X})^2$
198	$198 - 201 = -3$	$(-3)^2 = 9$
202	$202 - 201 = 1$	$(1)^2 = 1$
205	$205 - 201 = 4$	$(4)^2 = 16$
197	$197 - 201 = -4$	$(-4)^2 = 16$
200	$200 - 201 = -1$	$(-1)^2 = 1$
204	$204 - 201 = 3$	$(3)^2 = 9$

$$s = \sqrt{\frac{9 + 1 + 16 + 16 + 1 + 9}{6 - 1}}$$

$$s = \sqrt{\frac{52}{5}} = \sqrt{10.4} \approx 3.22$$

Hypotheses testing Example 1

Situation 1: One Sample t-Test - Example

Step 3: Compute the t-Statistic

The formula for the **one-sample t-statistic** is:

$$t = \frac{\bar{X} - \mu}{s/\sqrt{n}}$$

Substituting the values:

$$t = \frac{201 - 200}{3.22/\sqrt{6}}$$

$$t = \frac{1}{3.22/2.449}$$

$$t = \frac{1}{1.316} \approx 0.76$$

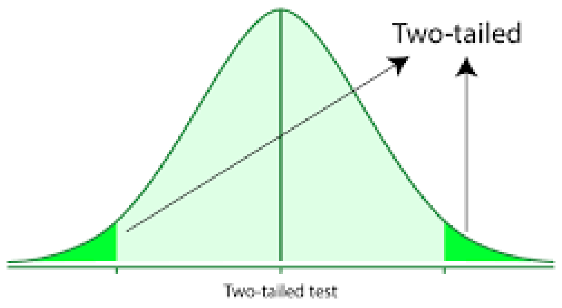
Hypotheses testing Example 1

Situation 1: **One Sample t-Test**

Step 4: Find the Critical t-Value

- Degrees of freedom: $df = n - 1 = 6 - 1 = 5.$
- Using a t-table for α = 0.05 (two-tailed test, df = 5):

$$t_{\text{critical}} = 2.571$$



df \ α	0.25	0.20	0.15	0.10	0.05	0.025	0.02	0.01	0.005	0.001
1	1.000	1.376	1.963	3.078	6.314	12.71	15.89	31.82	63.66	318.3
2	0.816	1.061	1.386	1.886	2.920	4.303	4.849	6.965	9.925	22.33
3	0.765	0.978	1.250	1.638	2.353	3.182	3.482	4.541	5.841	10.21
4	0.741	0.941	1.190	1.533	2.132	2.776	2.999	3.747	4.604	7.173
5	0.727	0.920	1.156	1.476	2.015	2.571	2.757	3.365	4.032	5.893
6	0.718	0.906	1.134	1.440	1.943	2.447	2.612	3.143	3.707	5.208
7	0.711	0.896	1.119	1.415	1.895	2.365	2.517	2.998	3.499	4.785
8	0.706	0.889	1.108	1.397	1.860	2.306	2.449	2.896	3.355	4.501
9	0.703	0.883	1.100	1.383	1.833	2.262	2.398	2.821	3.250	4.297
10	0.700	0.879	1.093	1.372	1.812	2.228	2.359	2.764	3.169	4.144
11	0.697	0.876	1.088	1.363	1.796	2.201	2.328	2.718	3.106	4.025
12	0.695	0.873	1.083	1.356	1.782	2.179	2.303	2.681	3.055	3.930
13	0.694	0.870	1.079	1.350	1.771	2.160	2.282	2.650	3.012	3.852
14	0.692	0.868	1.076	1.345	1.761	2.145	2.264	2.624	2.977	3.787

Hypotheses testing Example 1

Situation 1: One Sample t-Test

Step 5: Compare t-Statistic with Critical Value

- Calculated t = 0.76
- Critical t = ±2.571

$$|0.76| < 2.571$$

Since the calculated t-value is **less than** the critical value, we **fail to reject H_0.**

Hypotheses testing Example 1

Situation 1: One Sample t-Test

Step 6: Conclusion

At $\alpha = 0.05$, we **fail to reject the null hypothesis** because the test statistic is **not extreme enough**.

🔷 **Interpretation**: The sample provides **no strong evidence** that the average apple weight differs from 200g.

Hypotheses testing Example 1 (R)

Situation 1: One Sample t-Test

Code :

```
# Sample data: Apple weights
apple_weights <- c(198, 202, 205, 197, 200, 204)

# Perform a one-sample t-test against the population mean (200g)
t_test_result <- t.test(apple_weights, mu = 200)

# Print the results
print(t_test_result)
```

Hypotheses testing Example 1 (R)

Situation 1: One Sample t-Test

Output :

```
	One Sample t-test

data:  apple_weights
t = 0.76, df = 5, p-value = 0.478
alternative hypothesis: true mean is not equal to 200
95 percent confidence interval:
 198.18 203.82
sample estimates:
mean of x
    201
```

Hypotheses testing Example 1 (R)

Situation 1: One Sample t-Test

Interpretation :

- **t-statistic = 0.76** (matches our hand calculation ✅)
- **Degrees of freedom (df) = 5** (since $n - 1 = 6 - 1$)
- **p-value = 0.478**
  - **Since p > 0.05**, we **fail to reject H_0.**
  - There is **not enough evidence** to say the apples' weight differs from 200g.

Hypotheses testing Example 1 (SAS)

Situation 1: One Sample t-Test

Code :

```
/* Create the dataset */
DATA apples;
    INPUT weight @@;
    DATALINES;
    198 202 205 197 200 204
    ;
RUN;

proc print data=apples;
run;

/* Perform one-sample t-test */
PROC TTEST DATA=apples H0=200;
    VAR weight;
RUN;
```

Hypotheses testing Example 1 (SAS)

Situation 1: One Sample t-Test

Output :

Obs	weight
1	198
2	202
3	205
4	197
5	200
6	204

DF	t Value	Pr > |t|
5	0.76	0.4818

Hypotheses testing Example 2

Situation 2: Two Sample t-Test (Independent Samples)

🔷 Scenario: Testing if Website A Has Higher Engagement than Website B

A company tests **two website designs (A & B)** and wants to compare the **average time spent on the website** by visitors.

We collect **6 samples from each group**:

Website A (seconds)	Website B (seconds)
5.2	4.2
5.1	4.3
4.9	4.5
5.3	4.1
5.0	4.6
4.8	4.4

Hypotheses testing Example 2

Situation 2: Two Sample t-Test (Independent Samples)

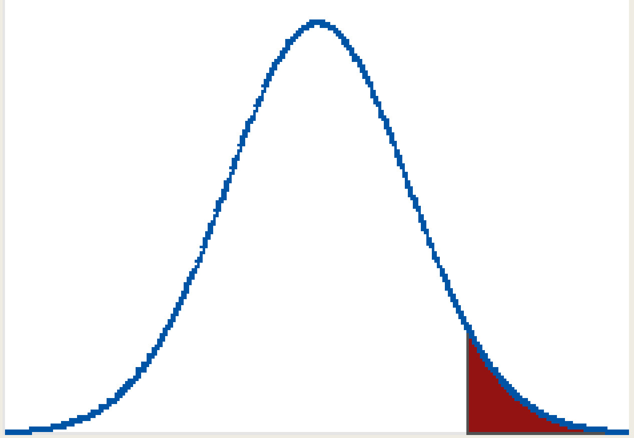
- **H_0 (Null Hypothesis):** The mean time spent on Website A is **less than or equal to** Website B

$$H_0 : \mu_A \leq \mu_B$$

- **H_1 (Alternative Hypothesis):** The mean time spent on Website A is **greater than** Website B

$$H_1 : \mu_A > \mu_B$$

This is a **right-tailed test**.



Hypotheses testing Example 2

Situation 2: Two Sample t-Test (Independent Samples)

🔷 Step 1: Compute Sample Statistics

Step 1.1: Calculate the Sample Means ($\bar{X}_A, \bar{X}_B$)

The formula for the sample mean:

$$\bar{X} = \frac{\sum X_i}{n}$$

For **Website A**:

$$\bar{X}_A = \frac{5.2 + 5.1 + 4.9 + 5.3 + 5.0 + 4.8}{6} = \frac{30.3}{6} = 5.05$$

For **Website B**:

$$\bar{X}_B = \frac{4.2 + 4.3 + 4.5 + 4.1 + 4.6 + 4.4}{6} = \frac{26.1}{6} = 4.35$$

Hypotheses testing Example 2

Situation 2: Two Sample t-Test (Independent Samples)

Step 1.2: Compute the Sample Standard Deviations (s_A, s_B)

The formula for **sample standard deviation:**

$$s = \sqrt{\frac{\sum(X_i - \bar{X})^2}{n-1}}$$

For Website A

X_A	$X_A - \bar{X}_A$	$(X_A - \bar{X}_A)^2$
5.2	0.15	0.0225
5.1	0.05	0.0025
4.9	-0.15	0.0225
5.3	0.25	0.0625
5.0	-0.05	0.0025
4.8	-0.25	0.0625

$$s_A = \sqrt{\frac{0.175}{5}} = \sqrt{0.035} \approx 0.187$$

For Website B

X_B	$X_B - \bar{X}_B$	$(X_B - \bar{X}_B)^2$
4.2	-0.15	0.0225
4.3	-0.05	0.0025
4.5	0.15	0.0225
4.1	-0.25	0.0625
4.6	0.25	0.0625
4.4	0.05	0.0025

$$s_B = \sqrt{\frac{0.175}{5}} = \sqrt{0.035} \approx 0.187$$

Hypotheses testing Example 2

Situation 2: Two Sample t-Test (Independent Samples)

Step 2: Compute the t-Statistic

The **pooled variance** (assuming equal variances):

$$s_p^2 = \frac{(n_A - 1)s_A^2 + (n_B - 1)s_B^2}{n_A + n_B - 2}$$

$$s_p^2 = \frac{(6-1)(0.187)^2 + (6-1)(0.187)^2}{6+6-2}$$

$$s_p^2 = \frac{5(0.035) + 5(0.035)}{10} = \frac{0.175 + 0.175}{10} = 0.035$$

$$s_p = \sqrt{0.035} \approx 0.187$$

The **t-statistic** formula:

$$t = \frac{\bar{X}_A - \bar{X}_B}{s_p \times \sqrt{\frac{1}{n_A} + \frac{1}{n_B}}}$$

$$t = \frac{5.05 - 4.35}{0.187 \times \sqrt{\frac{1}{6} + \frac{1}{6}}}$$

$$t = \frac{0.70}{0.187 \times \sqrt{0.333}}$$

$$t = \frac{0.70}{0.187 \times 0.577} = \frac{0.70}{0.108} \approx 6.48$$

Hypotheses testing Example 2

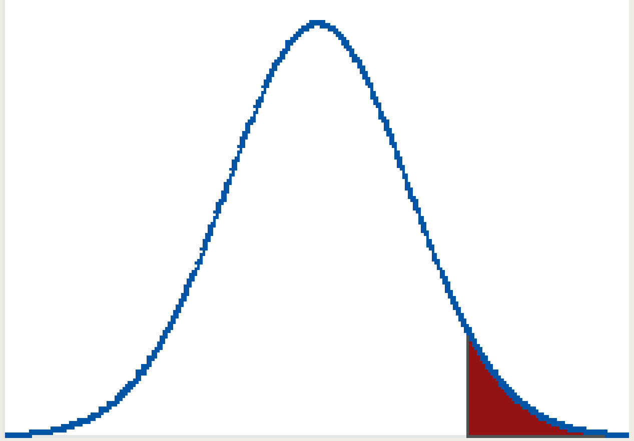
Situation 2: Two Sample t-Test (Independent Samples)

Step 3: Find the Critical t-Value

- **Degrees of freedom:**

$$df = n_A + n_B - 2 = 6 + 6 - 2 = 10$$

- **Significance level ($\alpha = 0.05$, one-tailed)**
  - From the **t-table**, the **critical t-value** for **df = 10, α = 0.05 (right-tailed)** is **1.812**.



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13	0.694	0.870	1.079	1.350	1.771	2.160	2.282	2.650	3.012	3.852
14	0.692	0.868	1.076	1.345	1.761	2.145	2.264	2.624	2.977	3.787

Hypotheses testing Example 2

Situation 2: Two Sample t-Test (Independent Samples)

Step 4: Compare t-Statistic with Critical t-Value

$$t_{\text{calculated}} = 6.48$$

$$t_{\text{critical}} = 1.812$$

Since **6.48 > 1.812**, we **reject the null hypothesis (H_0)**.

📌 **Step 5: Conclusion**

Since the **calculated t-statistic (6.48)** is **greater** than the **critical t-value (1.812)**, we **reject** H_0.

✅ **There is significant evidence that Website A has a higher engagement time than Website B at** α = **0.05.**

Hypotheses testing Example 2 (R)

Situation 2: Two Sample t-Test (Independent Samples)

Code :

```
# Sample data: Engagement time for Website A and Website B
website_A <- c(5.2, 5.1, 4.9, 5.3, 5.0, 4.8)
website_B <- c(4.2, 4.3, 4.5, 4.1, 4.6, 4.4)

# Perform a one-tailed (greater) two-sample t-test
t_test_result <- t.test(website_A, website_B, alternative = "greater", var.equal = TRUE)

# Print the results for the Pooled (Equal Variance) t-Test
print(t_test_result)
```

Hypotheses testing Example 2 (R)

Situation 2: Two Sample t-Test (Independent Samples)

Output :

```
        Two Sample t-test

data:  website_A and website_B
t = 6.4807, df = 10, p-value = 3.533e-05
alternative hypothesis: true difference in means is greater than 0
95 percent confidence interval:
 0.5042318       Inf
sample estimates:
mean of x mean of y 
     5.05      4.35 
```

Hypotheses testing Example 2 (R)

Situation 2: Two Sample t-Test (Independent Samples)

Interpretation :

🔷 **Key Observations from Your R Output**

- **t-Statistic: 6.4807** (matches our hand calculations and SAS results ✅)
- **Degrees of Freedom (df): 10** (same as SAS ✅)
- **p-value: 3.533e-05** (very small, meaning strong evidence to reject H_0 ✅)

📌 **Final Conclusion**

- Since **$p < 0.05$**, we **reject H_0**.
- Website A has **significantly higher** user engagement than Website B.

Hypotheses testing Example 2 (SAS)

Situation 2: Two Sample t-Test (Independent Samples)

Code :

```
/* Create the dataset */
DATA website_data;
    INPUT Group $ TimeSpent @@;
    DATALINES;
    A 5.2  A 5.1  A 4.9  A 5.3  A 5.0  A 4.8
    B 4.2  B 4.3  B 4.5  B 4.1  B 4.6  B 4.4
    ;
RUN;

proc print data=website_data;
run;

/* Perform one-tailed two-sample t-test */
PROC TTEST DATA=website_data SIDES=U;
    CLASS Group;
    VAR TimeSpent;
RUN;
```

Hypotheses testing Example 2 (SAS)

Situation 2: Two Sample t-Test (Independent Samples)

Output :

Method	Variances	DF	t Value	Pr > t
Pooled	Equal	10	6.48	<.0001
Satterthwaite	Unequal	10	6.48	<.0001

Hypotheses testing Example 2 (SAS)

Situation 2: Two Sample t-Test (Independent Samples)

Interpretation :

🔷 **Key Takeaways from Your SAS Output**

- **Both methods give the same t-value (6.48)** → Variance difference is small.
- **DF = 10 in both cases** → Suggests similar variances.
- **p-value < .0001** in both tests → Strong evidence that Website A has higher engagement.

Hypotheses testing Example 3

Situation 3: Paired t-Test (Dependent Samples)

Scenario: Testing if a New Training Program Decreases Sprint Time

A sports scientist wants to determine whether a **new sprint training program** helps athletes run **faster**. The scientist records the **100-meter sprint times** (in seconds) of **8 athletes** before and after they complete a **4-week specialized training program**.

Since lower sprint times indicate better performance, we are interested in testing whether the training program **reduces sprint time**, which corresponds to a **left-tailed paired t-test**.

Data Collection:

Each athlete's sprint time is measured **before and after** the training program.

Athlete	Before Training (seconds)	After Training (seconds)	Difference (D = Before - After)
1	11.2	10.8	0.4
2	10.9	10.5	0.4
3	11.5	11.1	0.4
4	12.0	11.4	0.6
5	11.8	11.3	0.5
6	10.7	10.4	0.3
7	11.0	10.6	0.4
8	12.2	11.8	0.4

Hypotheses testing Example 3

Situation 3: Paired t-Test (Dependent Samples)

Hypotheses for the Paired t-Test:

- **Null Hypothesis (H_0)**: The training program has no effect or increases sprint time.

$$H_0 : \mu_D \geq 0$$

where μ_D is the mean difference ($D = \text{Before} - \text{After}$).

- **Alternative Hypothesis (H_A)**: The training program **decreases** sprint time.

$$H_A : \mu_D < 0$$

Hypotheses testing Example 3

Situation 3: **Paired t-Test (Dependent Samples)**

Calculate the difference (D) between each pair of observations:

$$D_i = \text{Before Training Time} - \text{After Training Time}$$

Using the dataset:

Athlete	Before Training (seconds)	After Training (seconds)	Difference D (Before - After)
1	11.2	10.8	0.4
2	10.9	10.5	0.4
3	11.5	11.1	0.4
4	12.0	11.4	0.6
5	11.8	11.3	0.5
6	10.7	10.4	0.3
7	11.0	10.6	0.4
8	12.2	11.8	0.4

Hypotheses testing Example 3

Situation 3: **Paired t-Test (Dependent Samples)**

The given differences (D_i):

$$D = [0.4, 0.4, 0.4, 0.6, 0.5, 0.3, 0.4, 0.4]$$

Sum of the differences:

$$\sum D_i = 0.4 + 0.4 + 0.4 + 0.6 + 0.5 + 0.3 + 0.4 + 0.4 = 3.4$$

Now, compute the mean:

$$\bar{D} = \frac{\sum D_i}{n} = \frac{3.4}{8} = 0.425$$

$$\sum (D_i - \bar{D})^2 = 0.000625 + 0.000625 + 0.000625 + 0.030625 + 0.005625 + 0.015625 + 0.000625 + 0.000625 = 0.055$$

Now, compute the sample variance:

$$s_D^2 = \frac{\sum (D_i - \bar{D})^2}{n - 1} = \frac{0.055}{7} = 0.007857$$

Take the square root to get the standard deviation:

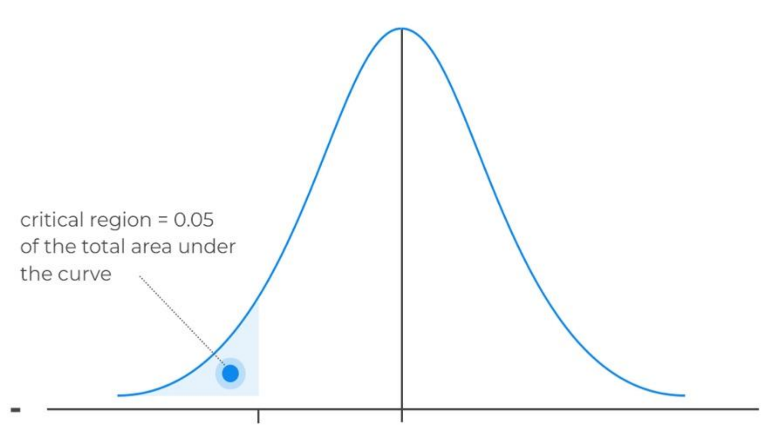
$$s_D = \sqrt{0.007857} \approx 0.0886$$

Hypotheses testing Example 3

Situation 3: Paired t-Test (Dependent Samples)

- **Degrees of freedom**: $df = n - 1 = 8 - 1 = 7$
- **Significance level**: $\alpha = 0.05$
- **Left-tailed t-critical value (from t-table for $df = 7$)**: $t_{\alpha,7} = -1.895$

Since $t = 13.57$ is **much larger than -1.895**, we **fail to reject** the null hypothesis H_0, meaning there is no sufficient evidence to conclude that the training program significantly reduces sprint time.



df \ α	0.25	0.20	0.15	0.10	0.05	0.025
1	1.000	1.376	1.963	3.078	6.314	12.71
2	0.816	1.061	1.386	1.886	2.920	4.303
3	0.765	0.978	1.250	1.638	2.353	3.182
4	0.741	0.941	1.190	1.533	2.132	2.776
5	0.727	0.920	1.156	1.476	2.015	2.571
6	0.718	0.906	1.134	1.440	1.943	2.447
7	0.711	0.896	1.119	1.415	1.895	2.365
8	0.706	0.889	1.108	1.397	1.860	2.306
9	0.703	0.883	1.100	1.383	1.833	2.262
10	0.700	0.879	1.093	1.372	1.812	2.228
11	0.697	0.876	1.088	1.363	1.796	2.201
12	0.695	0.873	1.083	1.356	1.782	2.179
13	0.694	0.870	1.079	1.350	1.771	2.160
14	0.692	0.868	1.076	1.345	1.761	2.145

Hypotheses testing Example 3 (R)

Situation 3: Paired t-Test (Dependent Samples)

Code :

```
#########################Paired T-Test###############
# Given data: Before and After sprint times
before <- c(11.2, 10.9, 11.5, 12.0, 11.8, 10.7, 11.0, 12.2)
after  <- c(10.8, 10.5, 11.1, 11.4, 11.3, 10.4, 10.6, 11.8)

#Perform built-in paired t-test (verification)
t.test(before, after, paired = TRUE, alternative = "less")
```

Hypotheses testing Example 3 (R)

Situation 3: Paired t-Test (Dependent Samples)

Output :

```
        Paired t-test

data:  before and after
t = 13.561, df = 7, p-value = 1
alternative hypothesis: true mean difference is less than 0
95 percent confidence interval:
      -Inf 0.4843745
sample estimates:
mean difference
          0.425
```

Hypotheses testing Example 3 (R)

Situation 3: **Paired t-Test (Dependent Samples)**

Interpretation :

1. **Test Statistic ($t = 13.561$)**:
   - This is a **very large positive t-value**, which suggests that the sample mean difference is much greater than 0.
   - Since this is a **left-tailed test**, a large positive t means we are in the opposite direction of the rejection region.
2. **p-Value = 1**:
   - The **p-value of 1** indicates that the probability of observing this data under the null hypothesis (H_0) is extremely high.
   - This means there is **no statistical evidence** to support the claim that the training program **reduces** sprint time.
3. **Conclusion:**
   - Since the **p-value > 0.05**, we **fail to reject** H_0.
   - This means we do **not have enough evidence** to conclude that the training program **reduces** sprint time.

Hypotheses testing Example 3 (SAS)

Situation 3: Paired t-Test (Dependent Samples)

Code :

```
DATA sprint_times;
    INPUT athlete before after;
    DIFF = before - after;  /* Compute the differences */
    DATALINES;
1 11.2 10.8
2 10.9 10.5
3 11.5 11.1
4 12.0 11.4
5 11.8 11.3
6 10.7 10.4
7 11.0 10.6
8 12.2 11.8
;
RUN;

proc print data=sprint_times;
run;


/* Step 2: Perform the Paired t-Test */

PROC TTEST DATA=sprint_times SIDES=L ALPHA=0.05;
    PAIRED before*after;
RUN;
```

Hypotheses testing Example 3 (SAS)

Situation 3: Paired t-Test (Dependent Samples)

Output :

DF	t Value	Pr < t
7	13.56	1.0000

Summary

- Step 1 : State the Hypotheses $(H_0 \ and \ H_A)$
- Step 2 : Set the Significance Level (α)
- Step 3 : Choose the correct test (t-test)
- Step 4 : Compute the Test Statistics & P Value
- Step 5 : Make a Conclusion (Reject or Fail to Reject H_0)

Type of Data	Test to Use	Example
1 sample vs. population mean	One-Sample t-Test	Test if a machine fills soda cans with 500ml on average
2 independent groups	Two-Sample t-Test	Test if men and women have different average heights
Same group before & after treatment	Paired t-Test ✅	Test if coffee reduces reaction time in the same individuals

Thank you

Malith Premarathna